

Customer Behavior Analysis

Python Code and PostegreSQL connection:

```
from sqlalchemy import create_engine
import pandas as pd

df=pd.read_csv(r"C:\Users\DELL\Downloads\customer_shopping_behavior.csv")

df['Review Rating']=df.groupby('Category')['Review Rating'].transform(lambda
x:x.fillna(x.median()))

df.columns=df.columns.str.lower()

df.columns=df.columns.str.replace(' ','_')

df=df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})

labels=['Young Adult','Adult','Middle Aged','Senior']

df['age_group']=pd.qcut(df['age'],q=4,labels=labels)

frequency_mapping={
    'Fortnightly':14,
    'Weekly':7,
    'Montly':30,
    'Quaterly':90,
    'Bi-Weekly':14,
    'Anually':365,
    'Every 3 Months':90
}

df['purchase_frequency_days']=df['frequency_of_purchases'].map(frequency_mapping)

df=df.drop('promo_code_used',axis=1)

username="postgres"

password="1981"

host="localhost"

port="5432"
```

```
database="customer_behaviour"
```

```
engine=create_engine(f"postgresql+psycopg2://{username}:{password}@{host}:{port}/{database}")
```

```
table_name="customer"
```

```
df.to_sql(table_name,engine,if_exists="replace",index=False)
```

```
print(f"data successfully loaded into '{table_name}' in database '{database}'.")
```

SQL CODE:

```
select gender,SUM(purchase_amount) AS revenue
```

```
from customer
```

```
group by gender;
```

	gender text	revenue numeric
1	Female	75191
2	Male	157890

```
select customer_id, purchase_amount
```

```
from customer
```

```
where discount_applied='Yes' and purchase_amount >= (select AVG(purchase_amount) from customer)
```

	customer_id bigint	purchase_amount bigint
81	163	74
82	164	67
83	165	65
84	166	60
85	169	80
86	173	67
87	176	86
88	177	81
89	181	81
90	182	61
91	183	96
92	188	94
93	190	82
94	191	70
95	192	76
96	194	100
97	195	94
98	197	88
99	198	78
100	200	73
101	201	61

```

select item_purchased,ROUND(AVG(review_rating::numeric),2) AS "Average product rating"
from customer
group by item_purchased
order by avg(review_rating) desc
limit 5;

```

	item_purchased text	Average product rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

```

select shipping_type,
ROUND(AVG(purchase_amount),2)
from customer
where shipping_type in ('Standard','Express')
group by shipping_type;

```

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

```

select subscription_status,
COUNT(customer_id) as total_customers,
ROUND(AVG(purchase_amount),2) as average_spend,
ROUND(SUM(purchase_amount),2) as total_revenue
from customer
group by subscription_status
order by total_revenue,average_spend desc;

```

	subscription_status text	total_customers bigint	average_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

```

select item_purchased,
ROUND(100 * SUM(CASE WHEN discount_applied ='Yes' THEN 1 ELSE 0
END)/COUNT(*),2) as discount_rate

```

```

from customer
group by item_purchased
order by discount_rate desc
limit 5;

```

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

```

with customer_type as(
select customer_id,previous_purchases,
CASE
  when previous_purchases=1 THEN 'NEW'
  when previous_purchases BETWEEN 2 and 10 THEN 'Returning'
  else 'Loyal'
end as customer_segment
from customer
)
select customer_segment,count(*) as "NUMBER OF CUSTOMERS"
from customer_type
group by customer_segment

```

	customer_segment text	NUMBER OF CUSTOMERS bigint
1	Loyal	3116
2	NEW	83
3	Returning	701

```

WITH item_counts as(
select category,

```

```

item_purchased,
COUNT(customer_id) as total_orders,
ROW_NUMBER() OVER(partition by category order by count(customer_id)DESC) as
item_rank
from customer
group by category,item_purchased
)

```

```

select item_rank,category,item_purchased,total_orders
from item_counts
where item_rank<=3;

```

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163

```

select subscription_status,
count(customer_id) as repeat_buyers
from customer
where previous_purchases>5
group by subscription_status

```

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

```

select age_group,

```

SUM(purchase_amount) as total_revenue

from customer

group by age_group

order by total_revenue desc;

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle Aged	59197
3	Adult	55978
4	Senior	55763

Power BI Dashboard:



Customer Behavior Dashboard



Customer Behavior Dashboard

